

Sales and Operations Dashboard

Architecture and Metric Definition Document

1. Purpose and Scope

This document defines the architecture for the Sales and Operations Dashboard sprint. It gives one exact definition for every KPI, the data model behind those KPIs, the API contract every screen must use, the access rules for financial data and the layout each team member should build to. Every other task on this sprint follows what is written here, so nobody needs to guess a formula, a field name or an access rule while building.

The goal of the dashboard itself is to turn raw sales, order, inventory and receivable records into a small set of numbers and charts a manager can read in under 10 seconds, with every number traceable back to the records it came from.

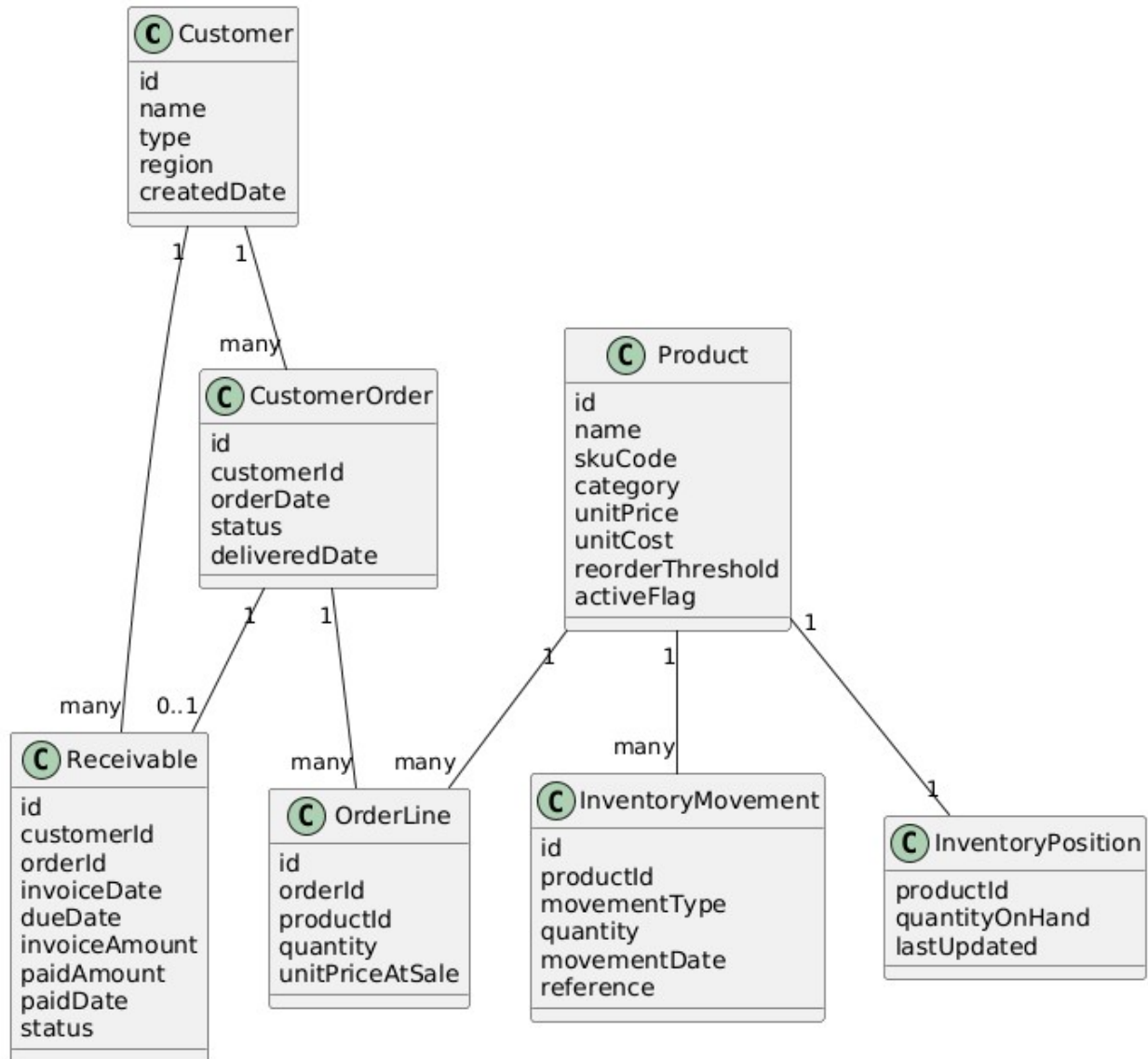
2. Data Architecture

2.1 Source Entities

These are the only tables the dashboard reads from. Every KPI in Section 3 is built from these fields and nothing else. Keeping one set of source tables is what guarantees the dashboard and the drill down screens always agree.

- Customer holds id, name, customer type, region, and created date.
- Product holds id, name, sku code, category, unit price, unit cost, reorder threshold, and active flag.
- CustomerOrder holds id, customer id, order date, status, and delivered date. Status is one of Pending, Processing, Shipped, Delivered, Cancelled, or Returned.
- OrderLine holds id, order id, product id, quantity, and unit price at the time of sale.
- InventoryMovement holds id, product id, movement type, quantity, movement date, and a reference such as an order id or a note. Movement type is one of Stock In, Stock Out, Adjustment, or Return.
- InventoryPosition holds product id, quantity on hand, and last updated date. It is a maintained snapshot, updated every time a movement is recorded, so current stock never needs to be recalculated from the full movement history.
- Receivable holds id, customer id, order id, invoice date, due date, invoice amount, paid amount, paid date, and status. Status is one of Unpaid, Partially Paid, or Paid.

2.2 Entity Relationship Diagram



2.3 Aggregation and Query Strategy

- Every KPI is computed directly from the source tables above using one indexed query per KPI group. No number is ever stored twice in two different formats.
- The same query logic used to compute a KPI value is reused to build its drill down list, so a card and its underlying list can never disagree.
- Where a KPI groups by month, product, or customer, that grouping happens inside the database query, not in application code, so it stays correct as data grows.
- For the demo scale dataset, direct queries against the source tables are enough. If a client dataset grows large, a nightly job can write daily totals into a small summary table, and the dashboard can read from that summary table for anything before today, while today itself is always computed live.

2.4 Financial Number and Date Handling

- All money fields are stored as whole numbers in the smallest currency unit, for example milligrams, never as decimals, so rounding errors cannot creep in. Rounding to a display value happens only in the frontend, never in storage or in a calculation.
- The whole demo uses a single currency unless a client explicitly asks for more than one.
- All dates and timestamps are stored in UTC. They convert to Pakistan Standard Time only when shown on screen.
- A date range filter is inclusive of the start date and covers up to and including the end date.
- Order date is the business date used for Sales, Orders, and Top Performer KPIs. Invoice date and due date are the business dates used for Receivables.
- When no range is chosen, the dashboard defaults to the current month. The standard presets are Today, This Week, This Month, This Quarter, This Year, and Custom Range.

2.5 Access Control for Financial Information

- Three roles exist. Admin and Manager can see every number and every drill down. Viewer can see aggregate counts and totals only, with no customer names, no unit cost, and no receivable detail.
- The role comes from the authenticated session on the server. It is never trusted from anything sent by the frontend.
- The full access matrix is in Section 6.

2.6 Seed and Import Validation

- Every order must reference a customer that already exists. Every order line must reference a product that already exists.
- Quantity on a normal order line must be a whole number greater than zero. A return line is the only case allowed to carry a negative quantity, and it must carry a return flag.
- Invoice numbers must be unique. Dates must be real calendar dates and cannot fall after the current system date.
- Any record that fails a rule above is rejected, not silently corrected. After every import, a short report lists how many records were accepted, how many were rejected, and why.

3. KPI Definitions and Formulas

Every KPI below has exactly one definition and one formula. Nobody should compute the same number a second way anywhere in the system.

3.1 Page 1, Sales Performance

Total Sales

Definition. The total value of everything sold in the selected date range.

Formula. Sum of quantity multiplied by unit price, across every order line, for orders that are not cancelled, where the order date falls in the selected range.

Total Orders

Definition. The number of orders placed in the selected date range.

Formula. Count of orders that are not cancelled, where the order date falls in the selected range.

Average Order Value

Definition. How much a typical order is worth.

Formula. Total Sales divided by Total Orders.

Sales Growth

Definition. How sales in this period compare to the period right before it.

Formula. Take Total Sales this period minus Total Sales in the previous period of equal length, divide by Total Sales in the previous period, and multiply by 100.

3.2 Page 2, Orders and Fulfillment

Orders Pending

Definition. Orders still waiting to be processed or shipped.

Formula. Count of orders with status Pending or Processing, where order date falls in the selected range.

Orders Delivered

Definition. Orders successfully completed in the range.

Formula. Count of orders with status Delivered, where order date falls in the selected range.

Orders Cancelled

Definition. Orders that did not go through.

Formula. Count of orders with status Cancelled, where order date falls in the selected range.

Fulfillment Rate

Definition. The share of placed orders that were actually delivered.

Formula. Orders Delivered divided by Total Orders placed in the same range, multiplied by 100.

3.3 Page 3, Inventory

Total Stock Value

Definition. The value of everything currently sitting in inventory. This is a live snapshot and is not affected by the date filter.

Formula. Sum of quantity on hand multiplied by unit cost, across all active products.

Items Low on Stock

Definition. Products that are close to running out.

Formula. Count of products where quantity on hand is greater than zero and less than or equal to that product's reorder threshold.

Items Out of Stock

Definition. Products with nothing left to sell.

Formula. Count of products where quantity on hand equals zero.

Total Active Products

Definition. How many products are currently sold.

Formula. Count of products marked active.

3.4 Page 4, Receivables

Total Outstanding

Definition. Money the business is still owed by customers right now. This is a live snapshot.

Formula. Sum of invoice amount minus paid amount, across every invoice with status not equal to Paid.

Overdue Amount

Definition. Money that is not just outstanding but already late.

Formula. Sum of invoice amount minus paid amount, for invoices where due date is before today and status is not Paid.

Overdue Invoices Count

Definition. How many separate invoices are late.

Formula. Count of invoices where due date is before today and status is not Paid.

Average Days to Pay

Definition. How long customers typically take to pay once invoiced.

Formula. Average of payment date minus invoice date, across invoices marked Paid within the selected range.

3.5 Page 5, Top Performers

Top Products by Sales

Definition. The five products bringing in the most revenue.

Formula. Rank every product by the sum of quantity multiplied by unit price across its order lines in the selected range, excluding cancelled orders. Show the top five.

Top Customers by Sales

Definition. The five customers bringing in the most revenue.

Formula. Rank every customer by the sum of their order totals in the selected range, excluding cancelled orders. Show the top five.

3.6 Page 6, Operational KPIs

Average Fulfillment Time

Definition. How many days it takes on average from order to delivery.

Formula. Average of delivered date minus order date, across orders with status Delivered in the selected range.

Return Rate

Definition. The share of orders that came back as returns.

Formula. Count of orders marked Returned divided by Total Orders in the same range, multiplied by 100.

Repeat Customer Rate

Definition. How many customers ordered more than once in the range.

Formula. Count of customers with more than one order in the selected range, divided by total unique customers in the same range, multiplied by 100.

Inventory Turnover

Definition. How efficiently stock is being sold and replaced.

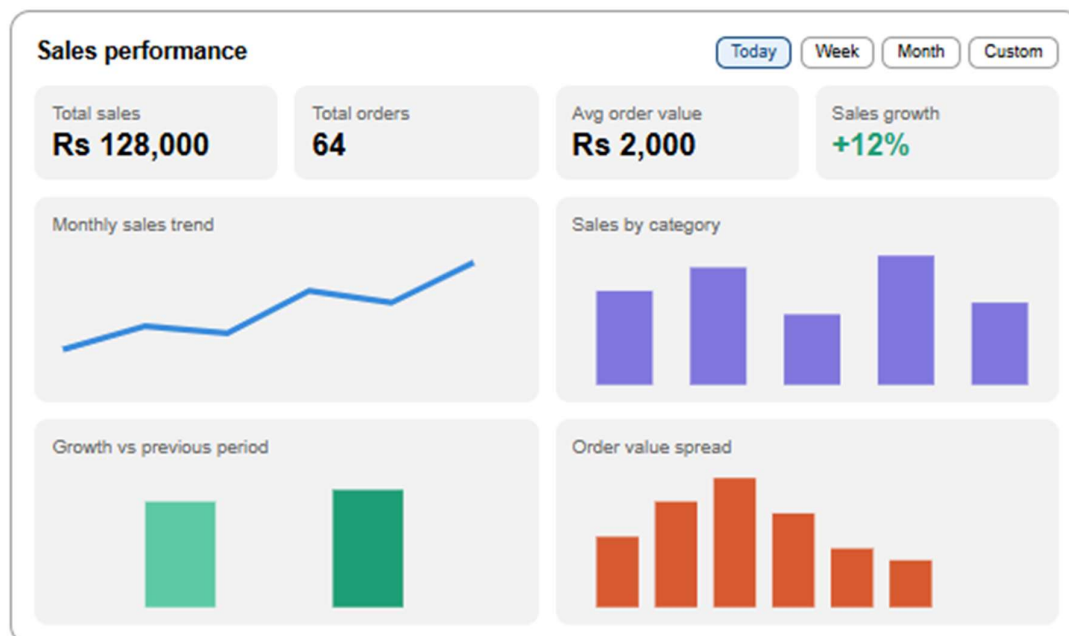
Formula. Cost of goods sold in the selected range divided by average inventory value in the same range.

4. Dashboard Pages and Layout

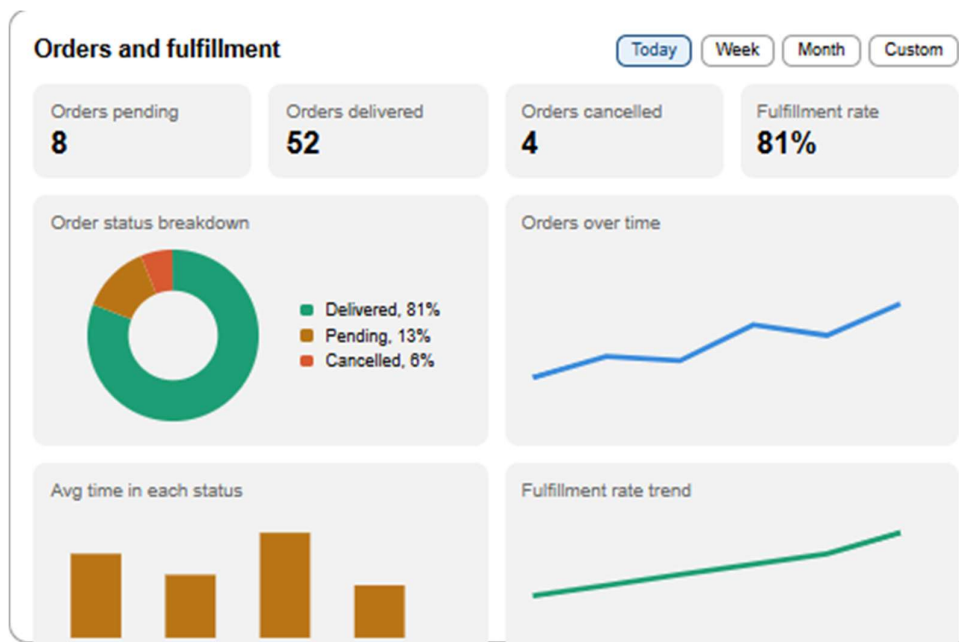
4.1 Layout Rules

- Each page shows three to four KPI cards plus 3 to 4 supporting chart or list. Nothing more.
- No page scrolls, not up and down, not left and right. Everything for that page fits in one screen, while ensuring readability.
- A date range control sits at the top of every page and affects every card and chart below it, except where a KPI is explicitly marked as a live snapshot in Section 3.
- KPI cards use large simple numbers and short plain labels, so the meaning is obvious without reading a legend.
- Every KPI card can be clicked to open the matching drill down list, filtered to exactly the records that produced that number.

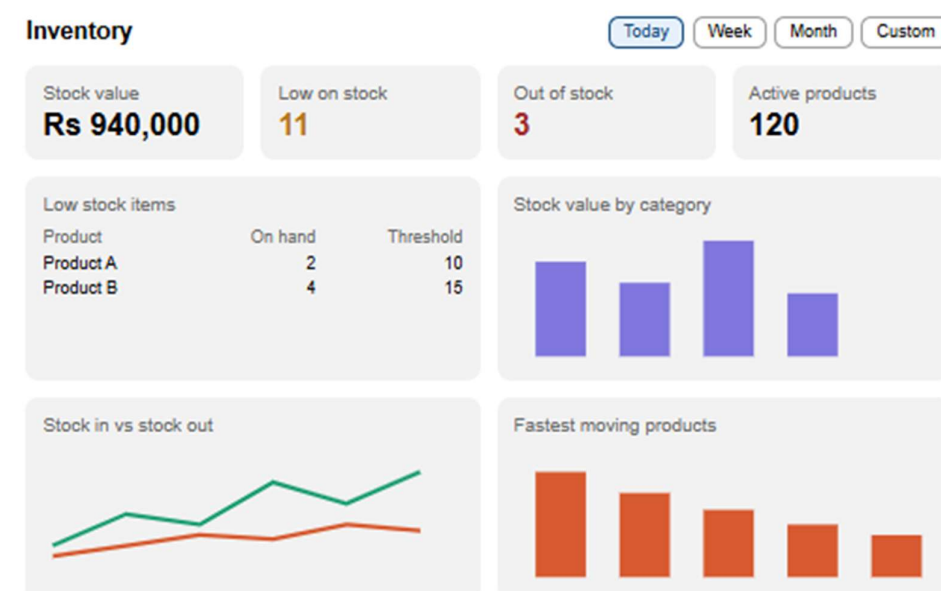
4.2 Page 1, Sales Performance



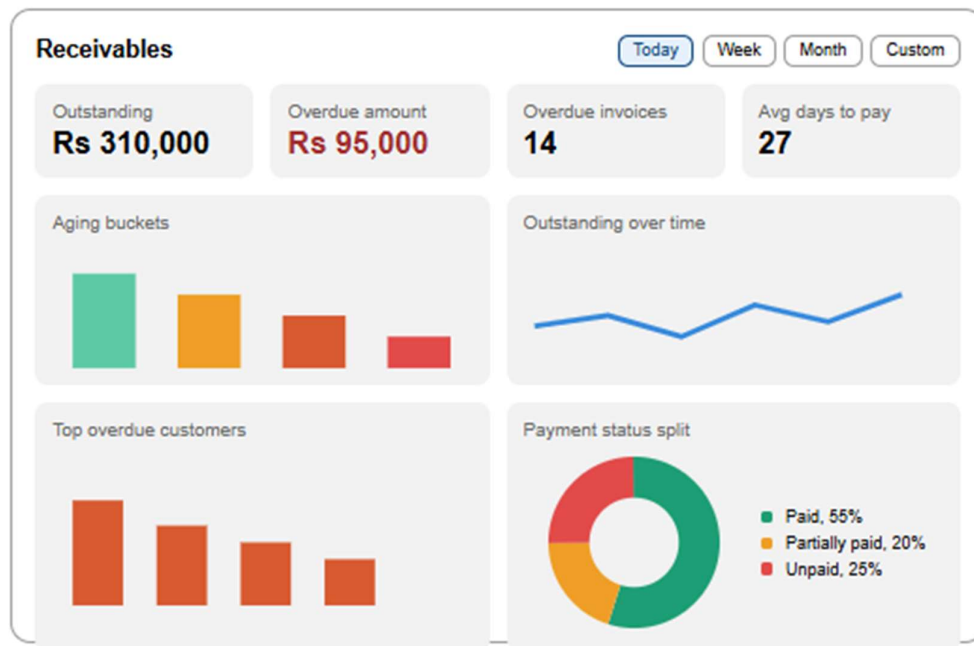
4.3 Page 2, Orders and Fulfillment



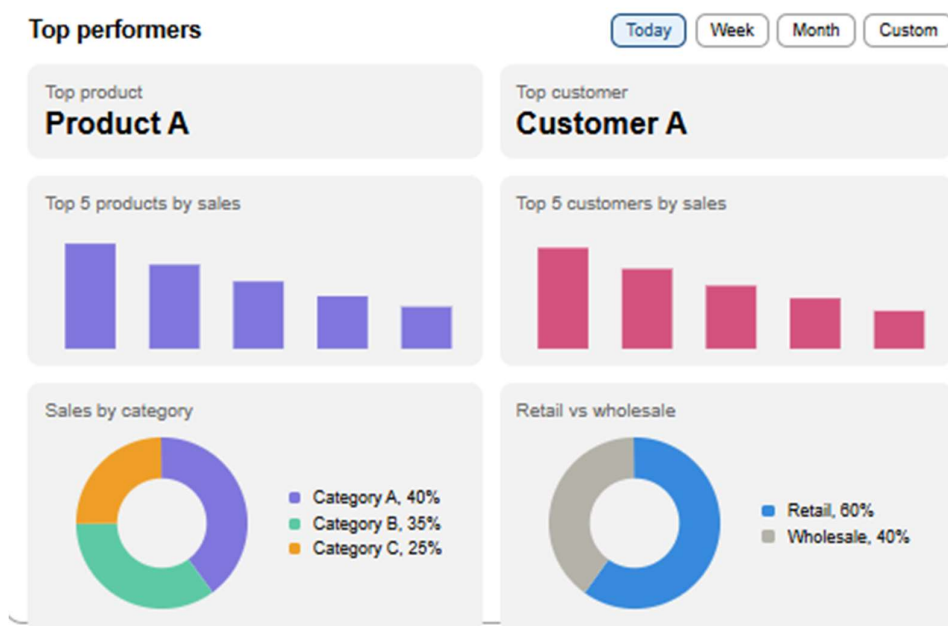
4.4 Page 3, Inventory



4.5 Page 4, Receivables



4.6 Page 5, Top Performers



4.7 Page 6, Operational KPIs

Operational KPIs

Today

Week

Month

Custom

Avg fulfillment time

3.2 days

Return rate

2%

Repeat customer rate

38%

Inventory turnover

4.1

Fulfillment time trend



Return rate trend



Repeat vs new customers



■ Repeat, 38%
■ New, 62%

Inventory turnover trend



5. API Contract

Every endpoint returns a recordCount field showing how many source records were used, and a generatedAt timestamp, so every number can be traced back to the records behind it. Every endpoint checks the caller's role before returning anything.

Method and Endpoint	Purpose	Key Parameters	Access
GET /api/dashboard/kpis/sales	Total Sales, Total Orders, Average Order Value, Sales Growth	from, to	All roles
GET /api/dashboard/trends/sales	Monthly sales series for the trend chart	from, to, granularity	All roles
GET /api/dashboard/kpis/orders	Orders Pending, Delivered, Cancelled, Fulfillment Rate	from, to	All roles
GET /api/dashboard/orders	Drill down order list	status, customer, from, to, page	Admin, Manager
GET /api/dashboard/orders/{id}	Single order detail	id	Admin, Manager
GET /api/dashboard/kpis/inventory/counts	Items Low on Stock, Out of Stock, Total Active Products	none	All roles
GET /api/dashboard/kpis/inventory/value	Total Stock Value	none	Admin, Manager
GET /api/dashboard/inventory/lowstock	Drill down low stock list	page	All roles
GET /api/dashboard/kpis/receivables	Total Outstanding, Overdue Amount, Overdue Invoices, Average Days to Pay	from, to	Admin, Manager
GET /api/dashboard/receivables/aging	Aging bucket breakdown	none	Admin, Manager
GET /api/dashboard/receivables/{id}	Single invoice detail	id	Admin, Manager
GET /api/dashboard/rankings/products	Top products by sales	from, to, limit	Admin, Manager
GET /api/dashboard/rankings/customers	Top customers by sales	from, to, limit	Admin, Manager
GET /api/dashboard/kpis/operational	Fulfillment Time, Return Rate, Repeat Customer	from, to	Admin, Manager

	Rate, Inventory Turnover		
--	-----------------------------	--	--

6. Security Matrix

This table is the single source of truth for who can see what. Every endpoint in Section 5 enforces exactly this.

Data Area	Admin	Manager	Viewer
Sales KPIs and Trend	Full	Full	Aggregate only
Orders and Status	Full	Full	Aggregate only
Inventory Counts	Full	Full	Aggregate only
Inventory Value and Unit Cost	Full	Full	No access
Receivables, Summary and Detail	Full	Full	No access
Top Products and Customers	Full	Full	No access
Drill Down to Order or Invoice Records	Full	Full	No access

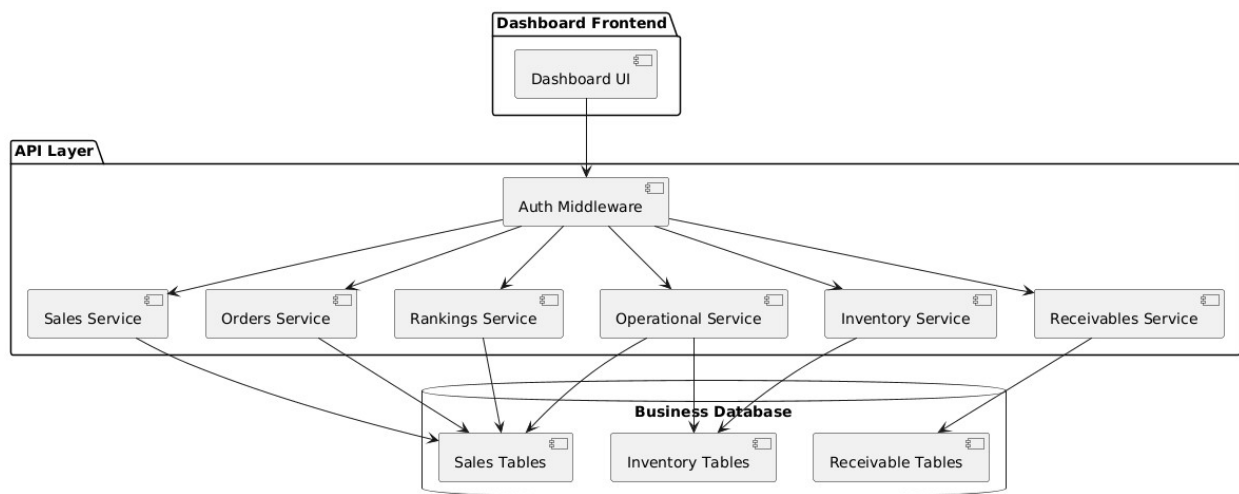
7. Performance Notes

- Add an index on order date, order status, and customer id for the CustomerOrder table, on order id and product id for OrderLine, on product id and movement date for InventoryMovement, and on due date and status for Receivable.
- Cache each KPI summary response for sixty seconds. The underlying data does not change fast enough during a demo or a normal working day to need anything faster.
- Every drill down list is paginated. Default page size is twenty five records.
- Compute related numbers together in one query rather than several round trips. For example, Total Sales, Total Orders, and Average Order Value come from a single query.
- The seed dataset is small enough that direct queries against the source tables are enough. If a real client dataset grows large, add a nightly job that writes daily totals into a summary table, and read from that summary table for anything before today.

8. Optional Addition, Insight Panel

A small panel that reads the same aggregate KPI numbers already shown on the dashboard, never raw customer or financial records, and writes one or two plain sentences summarizing the period. For example, noting that sales grew by a certain percent mainly because of one product, while also flagging that overdue invoices went up. The first version can be a simple rule that compares this period to the previous period and picks the two biggest movers. A real AI generated summary can replace that rule later using the exact same input numbers, so the security boundary in Section 6 never changes.

9. System Diagram



10. Team Responsibilities and Step by Step Plan

10.1 Who Owns What

Person	Owns
Haroon	Architecture, KPI formulas, API contract, security matrix, performance notes, final review
Hassan	Database schema, seed and import script, API endpoints, automated tests
Ibrahim	Realistic seed dataset, order and inventory and receivable drill down screens, edge case coverage
Zaid	Dashboard UI, KPI cards, charts, date filters, responsive layout

10.2 Step by Step Order of Work

- Step 1, Haroon. Share this document with the team before any code is written, so every formula and field name is agreed up front.
- Step 2, Hassan. Build the database schema from Section 2.1, and build the seed and import script following the validation rules in Section 2.6.
- Step 3, Ibrahim. Using Hassan's schema, build the seed dataset. Include at least one low stock product, one out of stock product, one overdue invoice, one cancelled order, and one product with zero sales, so every edge case the sprint requires is present from day one.
- Step 4, Hassan. Build the API endpoints in Section 5, one KPI group at a time, in this order, Sales, then Orders, then Inventory, then Receivables, then Rankings, then Operational. Write one automated test per endpoint against Ibrahim's seed data with a known expected total.
- Step 5, Zaid. Build the six dashboard pages in the order given in Section 4, wiring each KPI card and chart to its matching endpoint as soon as Hassan finishes it, rather than waiting for everything at once.
- Step 6, Ibrahim. Build the order list, inventory list, and receivable list drill down screens, linked from the matching KPI cards on Zaid's pages.
- Step 7, All. Manually check a handful of dashboard numbers by hand against the raw seed data, to confirm every formula in Section 3 was built exactly as written.
- Step 8, All. Demonstrate the date range filter changing every relevant number, and demonstrate at least one drill down from a KPI card into its underlying records.
- Step 9, Haroon. Final architecture review against the acceptance checklist in Section 11.

11. Acceptance Checklist

- Every KPI has exactly one definition, matching Section 3 word for word.
- Financial data is protected exactly as written in Section 6, with no exceptions made during a demo.
- No number is hard coded anywhere in the UI. Every number on screen comes from a live API call.
- Every date filter change produces a new, correct result on every affected card and chart.
- Cancelled and returned records are handled the same way everywhere they appear.
- Queries behave the same way every time they run against the same date range and the same data.